# Homework assignment for 02YELMA
# Chapter: Electric Fields in Matter

1. An infinitely long dielectric cylinder of radius $R$ is polarized as $\mathbf{P}(s) = \frac{a}{s}\,\hat{s}$ for $0 < r \leq R$, where $a$ is a constant and $s$ is the distance from the axis of the cylinder.

   (a) Find the bound volume charge density $\rho_b$.

   (b) Find the bound surface charge density $\sigma_b$ at $s = R$.

   (c) Compute the total bound charge per unit length.

   (d) Calculate the total charge and explain why the result is surprising (or not). How can there be a net surface charge if the divergence is zero everywhere?

2. A dielectric slab occupies the region $-L \leq x \leq L$ and is infinite in the $y$ and $z$ directions. The polarization is given by $\mathbf{P}(x) = kx^2\,\hat{x}$ where $k$ is a constant.

   (a) Find the bound volume charge density $\rho_b(x)$.

   (b) Find the bound surface charge densities at $x = +L$ and $x = -L$.

   (c) Compute the total bound charge per unit area.

   (d) Determine whether the system is overall neutral. How would you explain this result?

   (e) Where is charge concentrated more—near the center or near the boundaries? Explain why.

3. A solid sphere of radius $R$ has polarization $\mathbf{P}(r) = kr\,\hat{r}$.

   (a) Find $\rho_b$ and $\sigma_b$.

   (b) Calculate the electric field inside and outside the sphere.

   (c) Write the expression for the potential $V(0)$ at the center of the sphere.

   (d) Compute the volume contribution and the surface contribution.

   (e) Add them and simplify.

   (f) Explain why the two contributions do or do not cancel.